

```

root_path = fileparts(mfilename("fullpath"));
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit200', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
used_ds = containers.Map()

```

```
used_ds =
```

```
Map with properties:
```

```

    Count: 0
   KeyType: char
  ValueType: any

```

```

sel_info = struct('cuts',[],'en_range',[],'fname','','bg_par',[]);
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
sel_info.en_range = 30:10:110;
sel_info.fname='EnFit_Ei200ref001off110_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br110dir001')=sel_info;
%
sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
sel_info.en_range = 30:10:130;
sel_info.fname='EnFit_Ei200ref001off200_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei200Br200dir001')=sel_info;
%
sel_info.cuts = {cuts2fit200.cut200_sel};
sel_info.en_range = 40:10:140;
sel_info.fname='EnFit_Ei200Br200dir010_2peaks_2D';
sel_info.bg_par= {};
used_ds('Ei200Br200dir010_2peaks')=sel_info;
%
sel_info.cuts = {cut(cuts2fit200.cut200_sel,[0,0.02,1],[[]])};

```

```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 4690699 pixels -- processing data... -----> retained 4604422 pixels
-----
Number of points in input object: 8259853
Fraction of object selected: 56.79% (= 4690699 points) ; Access speed: 0.96GB/sec
Fraction of object retained: 55.74% (= 4604422 points)

**** Total time in cut(sqw) : 0.8sec Including:
Data access time fraction: 39.5%; Transf. time frac.: 38.1%; Resulting pix preparation: 22.3%
-----

```

```

sel_info.en_range = 40:10:140;
sel_info.fname='EnFit_Ei200Br200dir010_1peak_2D';
sel_info.bg_par= {};
used_ds('Ei200Br200dir010_1peak')=sel_info;
%
sel_info.cuts = {cuts2fit200.cut110_sel};
sel_info.en_range = 30:10:110;

```

```
sel_info.fname='EnFit_Ei200Br110dir010_2D';
sel_info.bg_par= {};
used_ds('Ei200Br110dir010')=sel_info;
```

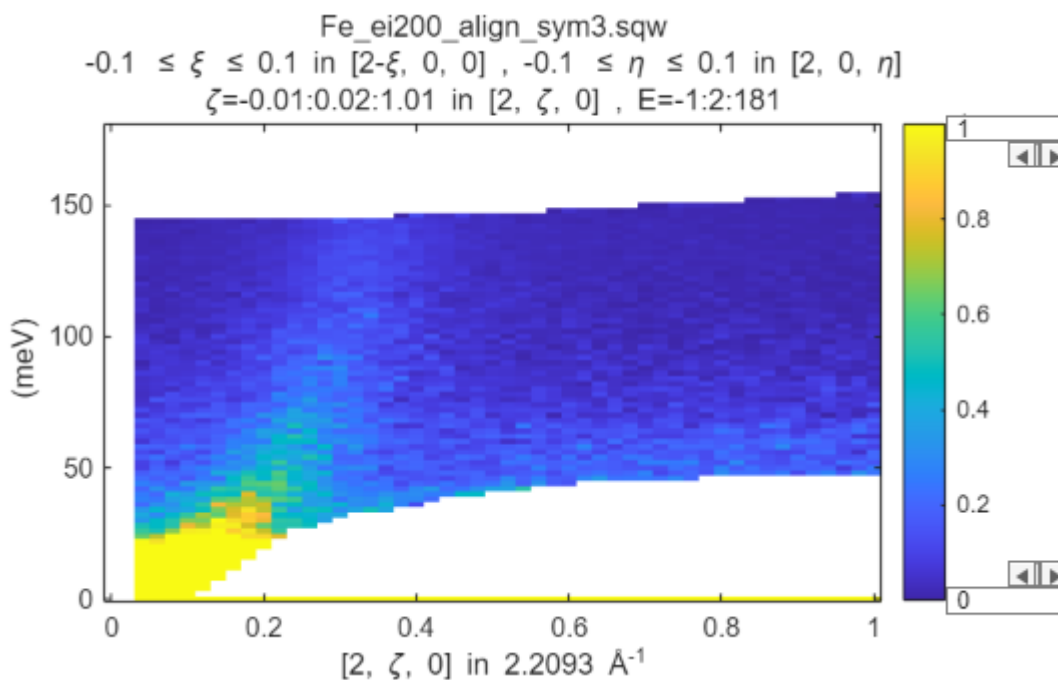
```
%cuts_list = cuts2fit200.cutsS_list();
%cuts_list = {cuts2fit200.cut200_200sel};
%cuts_list = {cuts2fit200.cut110_sel};
```

```
select= 'Ei200Br200dir010_1peak';
sel_info = used_ds(select);
```

```
cuts_list = sel_info.cuts
```

```
cuts_list = 1×1 cell array
    {1×1 sqw}
```

```
mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end
```



```
dir_name = "GH";
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10; % half width of data binning
%cut_en = 30:10:120;
%cut_en = 40:10:150;
%cut_en = 145;
```

```
cut_en = sel_info.en_range;

bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;
```

```
[fit_res_200fix_bg,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_en,dE_step,half_dE,true);
```

**** En = 40±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 182340 pixels -- processing data... -----> included 163713 pixels

Number of points in input object: 182340

Fraction of object selected: 100.00% (= 182340 points) ; Access speed: 1.98GB/sec

Fraction of object retained: 89.78% (= 163713 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 31.8%; Transf. time frac.: 65.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 182340 pixels -- processing data... -----> retained 163713 pixels

Number of points in input object: 182340

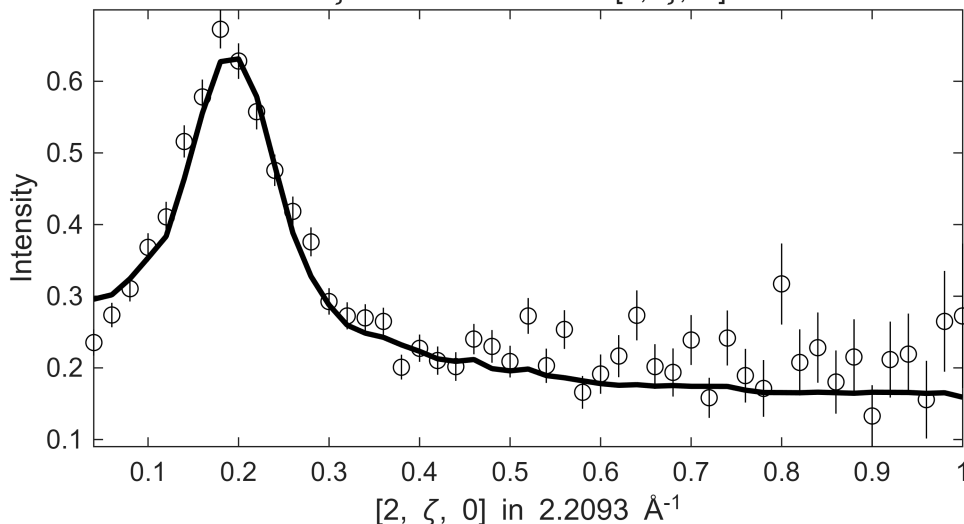
Fraction of object selected: 100.00% (= 182340 points) ; Access speed: 1.76GB/sec

Fraction of object retained: 89.78% (= 163713 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 22.6%; Transf. time frac.: 56.6%; Resulting pix preparation: 20.8%

Fe_ei200_align_sym3.sqw
S=0.54±0.036; J0= 24±0.34; gamma=3.78309±1.07322
-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 30 ≤ E ≤ 50
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 182340 pixels -- processing data... -----> retained 163713 pixels

Number of points in input object: 182340

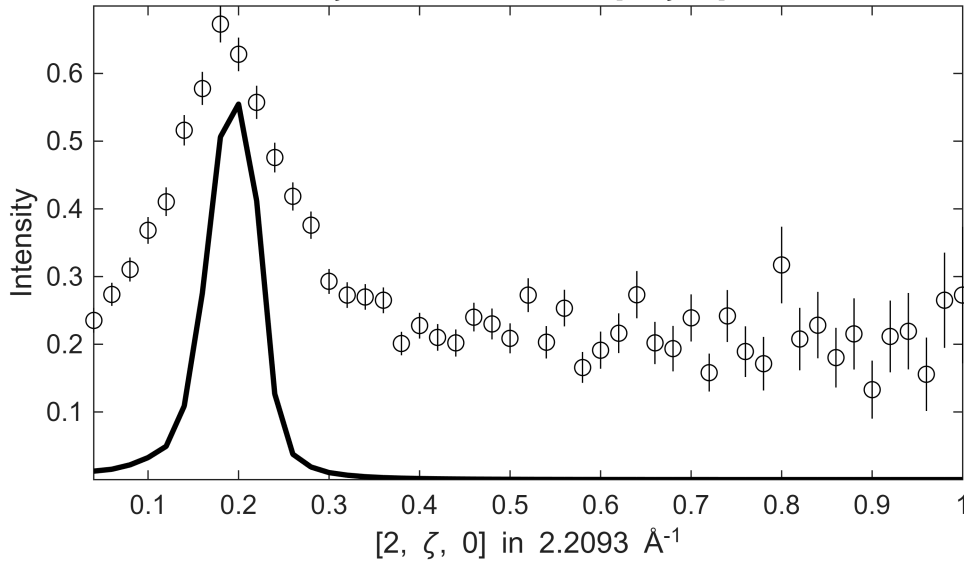
Fraction of object selected: 100.00% (= 182340 points) ; Access speed: 1.72GB/sec

Fraction of object retained: 89.78% (= 163713 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 24.3%; Transf. time frac.: 56.4%; Resulting pix preparation: 19.3%

Fe_ei200_align_sym3.sqw
-0.1 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, -0.1 $\leq \eta \leq 0.1$ in $[2, 0, \eta]$, 30 $\leq E \leq 50$
 $\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



**** En = 50±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 284010 pixels -- processing data... -----> included 261072 pixels

Number of points in input object: 284010

Fraction of object selected: 100.00% (= 284010 points) ; Access speed: 1.88GB/sec

Fraction of object retained: 91.92% (= 261072 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 39.8%; Transf. time frac.: 57.5%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 284010 pixels -- processing data... -----> retained 261072 pixels

Number of points in input object: 284010

Fraction of object selected: 100.00% (= 284010 points) ; Access speed: 1.96GB/sec

Fraction of object retained: 91.92% (= 261072 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 26.7%; Transf. time frac.: 51.9%; Resulting pix preparation: 21.4%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 284010 pixels -- processing data... -----> retained 261072 pixels

Number of points in input object: 284010

Fraction of object selected: 100.00% (= 284010 points) ; Access speed: 1.71GB/sec

Fraction of object retained: 91.92% (= 261072 points)

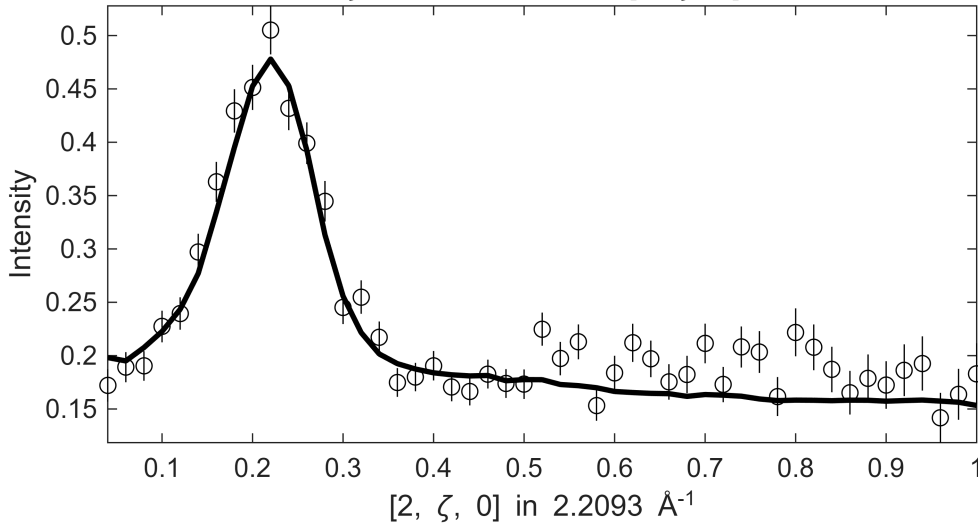
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 26.7%; Transf. time frac.: 56.8%; Resulting pix preparation: 16.5%

Fe_ei200_align_sym3.sqw

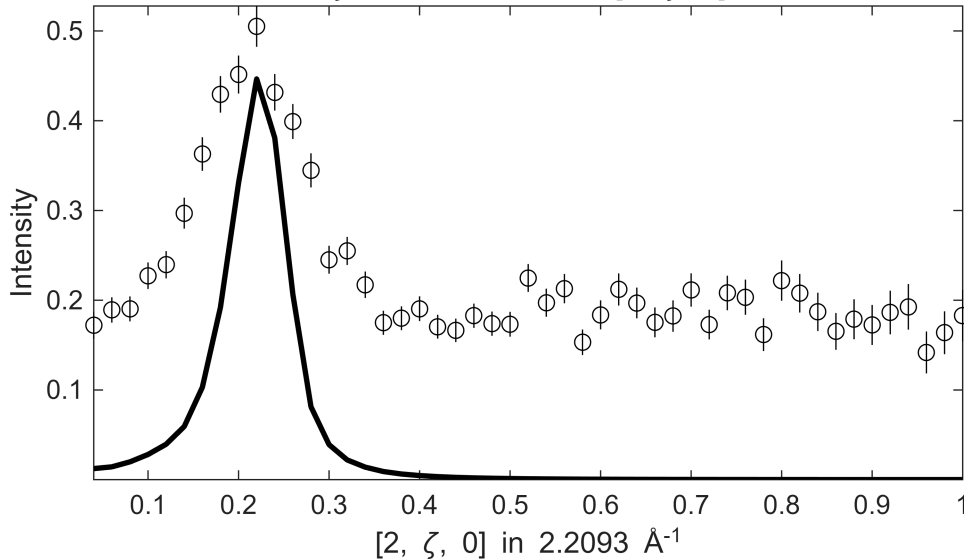
S=0.58±0.036; J0= 24±0.37; gamma=8.31812±1.4095

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 40 ≤ E ≤ 60
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 40 ≤ E ≤ 60
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



**** En = 60±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 342468 pixels -- processing data... -----> included 310966 pixels

Number of points in input object: 342468

Fraction of object selected: 100.00% (= 342468 points) ; Access speed: 1.94GB/sec

Fraction of object retained: 90.80% (= 310966 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 41.9%; Transf. time frac.: 55.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 342468 pixels -- processing data... -----> retained 310966 pixels

Number of points in input object: 342468

Fraction of object selected: 100.00% (= 342468 points) ; Access speed: 1.63GB/sec

Fraction of object retained: 90.80% (= 310966 points)

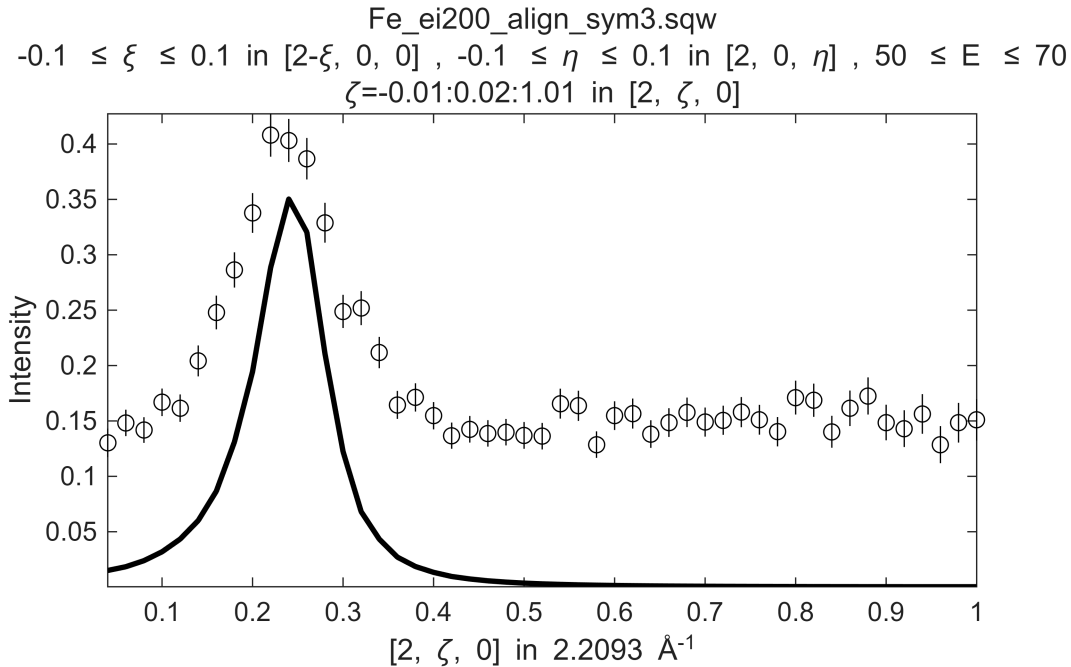
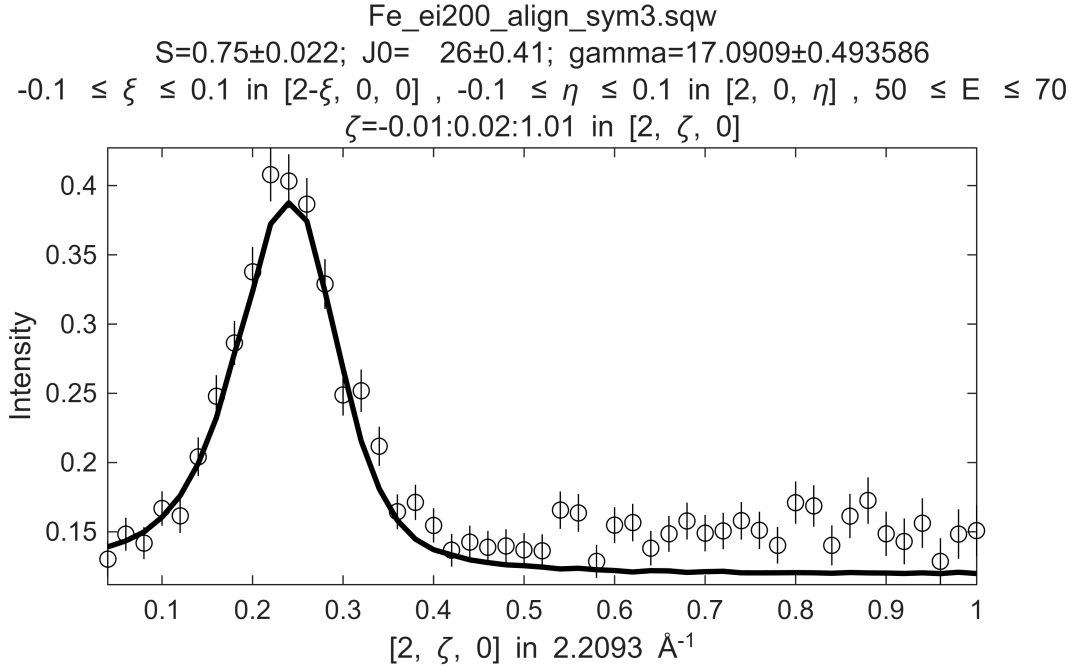
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 22.9%; Transf. time frac.: 45.1%; Resulting pix preparation: 32.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 342468 pixels -- processing data... -----> retained 310966 pixels

Number of points in input object: 342468
Fraction of object selected: 100.00% (= 342468 points) ; Access speed: 1.78GB/sec
Fraction of object retained: 90.80% (= 310966 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 26.0%; Transf. time frac.: 54.0%; Resulting pix preparation: 20.0%



**** En = 70±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 368343 pixels -- processing data... -----> included 334606 pixels

Number of points in input object: 368343
Fraction of object selected: 100.00% (= 368343 points) ; Access speed: 1.96GB/sec
Fraction of object retained: 90.84% (= 334606 points)

```

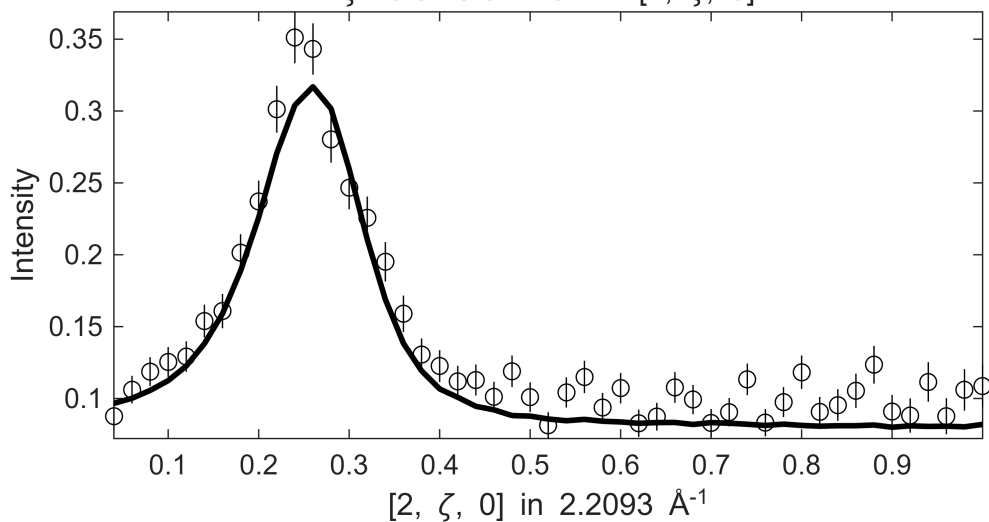
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction:  43.6%; Transf. time frac.:  54.0%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  368343 pixels -- processing data... -----> retained  334606 pixels
-----
Number of points in input object: 368343
      Fraction of object selected:  100.00% (=  368343 points) ; Access speed:      1.62GB/sec
      Fraction of object retained:   90.84% (=  334606 points)

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  28.4%; Transf. time frac.:  53.7%; Resulting pix preparation:  17.9%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  368343 pixels -- processing data... -----> retained  334606 pixels
-----
Number of points in input object: 368343
      Fraction of object selected:  100.00% (=  368343 points) ; Access speed:      1.64GB/sec
      Fraction of object retained:   90.84% (=  334606 points)

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  26.2%; Transf. time frac.:  55.3%; Resulting pix preparation:  18.4%
-----

```

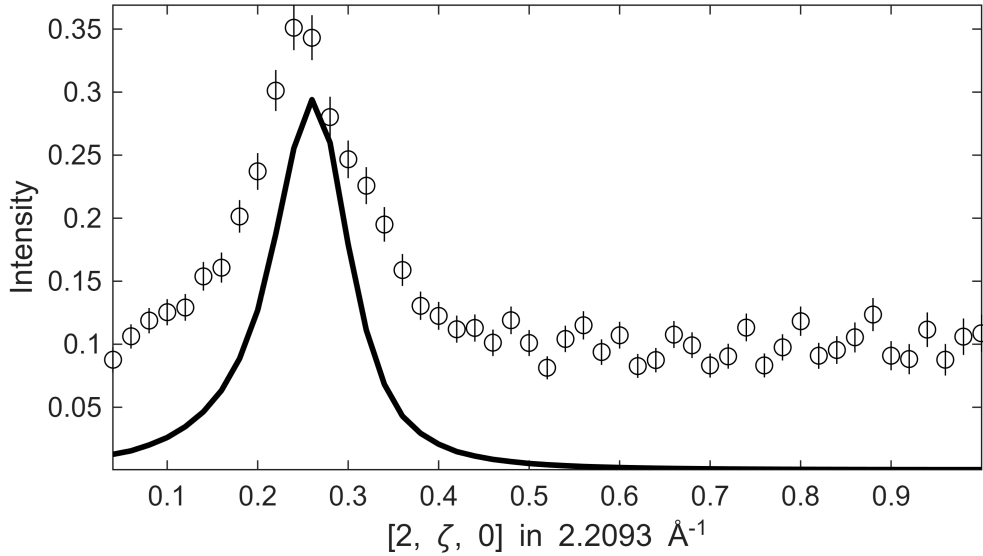
Fe_ei200_align_sym3.sqw
 $S=0.79\pm0.049$; $J_0=27\pm0.49$; $\gamma=22.6512\pm2.55952$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E \leq 80$ eV
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$



Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E \leq 80$

$\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



**** En = 80±10

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 404032 pixels -- processing data... -----> included 366804 pixels

Number of points in input object: 404032

Fraction of object selected: 100.00% (= 404032 points) ; Access speed: 1.94GB/sec

Fraction of object retained: 90.79% (= 366804 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 42.7%; Transf. time frac.: 55.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 404032 pixels -- processing data... -----> retained 366804 pixels

Number of points in input object: 404032

Fraction of object selected: 100.00% (= 404032 points) ; Access speed: 1.19GB/sec

Fraction of object retained: 90.79% (= 366804 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 37.8%; Transf. time frac.: 44.9%; Resulting pix preparation: 17.3%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 404032 pixels -- processing data... -----> retained 366804 pixels

Number of points in input object: 404032

Fraction of object selected: 100.00% (= 404032 points) ; Access speed: 1.70GB/sec

Fraction of object retained: 90.79% (= 366804 points)

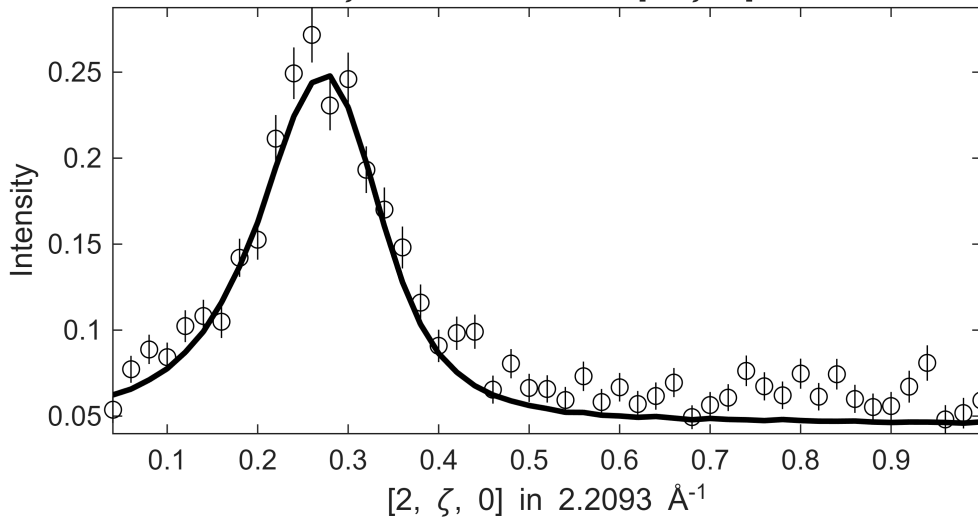
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 25.6%; Transf. time frac.: 50.3%; Resulting pix preparation: 24.1%

Fe_ei200_align_sym3.sqw

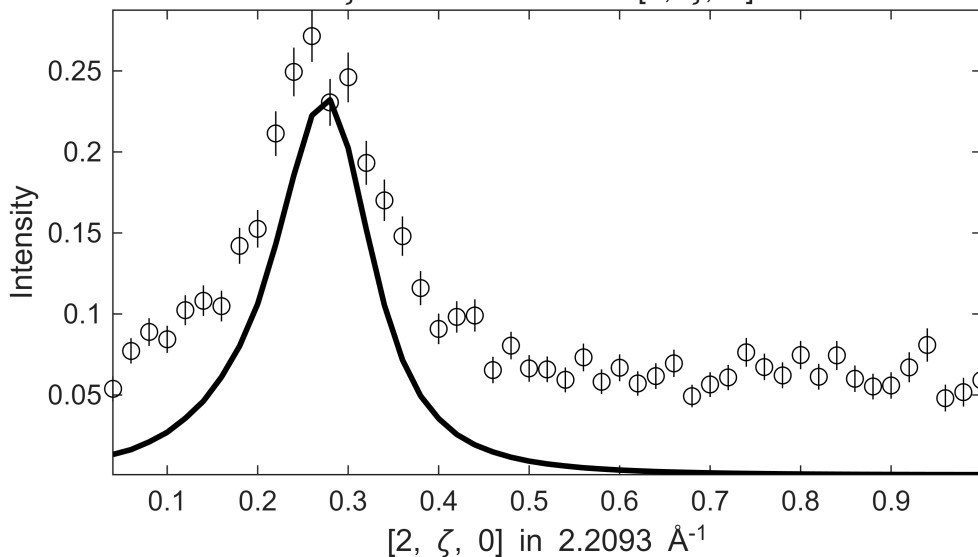
S=0.93±0.059; J0= 29±0.64; gamma=35.8787±3.74423

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 70 ≤ E ≤ 90
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 70 ≤ E ≤ 90
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



**** En = 90±10

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.71301

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 452915 pixels -- processing data... -----> included 411177 pixels

Number of points in input object: 452915

Fraction of object selected: 100.00% (= 452915 points) ; Access speed: 2.01GB/sec

Fraction of object retained: 90.78% (= 411177 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 43.8%; Transf. time frac.: 53.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; returning pixels - kept

*** Step 1 of 1; Read data for 449588 pixels -- processing data... -----> retained 408050 pixels

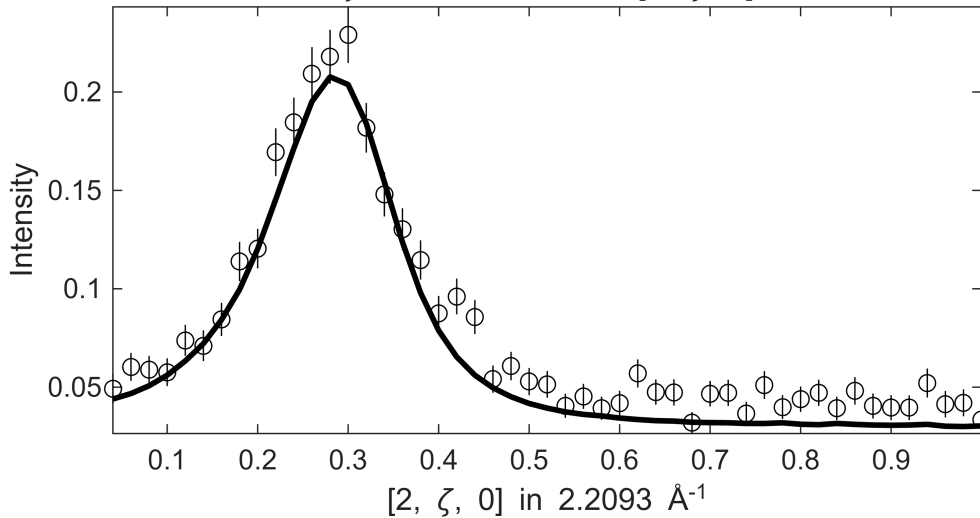
Number of points in input object: 449588

Fraction of object selected: 100.00% (= 449588 points) ; Access speed: 1.58GB/sec

Fraction of object retained: 90.76% (= 408050 points)

**** Total time in cut(sqws) : 0.1sec Including:
 Data access time fraction: 29.5%; Transf. time frac.: 52.5%; Resulting pix preparation: 18.0%

Fe_ei200_align_sym3.sqw
 $S=0.96\pm0.061$; $J0=31\pm0.71$; $\gamma=44.0336\pm4.45509$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 10$
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$

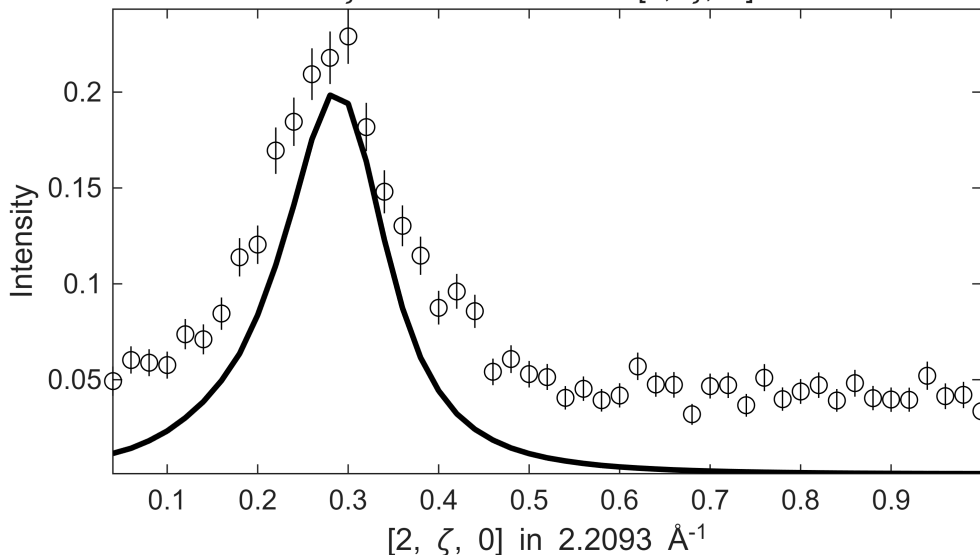


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 452915 pixels -- processing data... -----> retained 411177 pixels

Number of points in input object: 452915
 Fraction of object selected: 100.00% (= 452915 points) ; Access speed: 1.89GB/sec
 Fraction of object retained: 90.78% (= 411177 points)

**** Total time in cut(sqws) : 0.1sec Including:
 Data access time fraction: 27.1%; Transf. time frac.: 51.9%; Resulting pix preparation: 20.9%

Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 10$
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$



**** $E_n = 100\pm10$

Warning: Dataset:8 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.426

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 506911 pixels -- processing data... -----> included 460829 pixels

Number of points in input object: 506911

Fraction of object selected: 100.00% (= 506911 points) ; Access speed: 1.87GB/sec
Fraction of object retained: 90.91% (= 460829 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 46.9%; Transf. time frac.: 51.7%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 499694 pixels -- processing data... -----> retained 454198 pixels

Number of points in input object: 499694

Fraction of object selected: 100.00% (= 499694 points) ; Access speed: 1.46GB/sec
Fraction of object retained: 90.90% (= 454198 points)

**** Total time in cut(sqw) : 0.1sec Including:

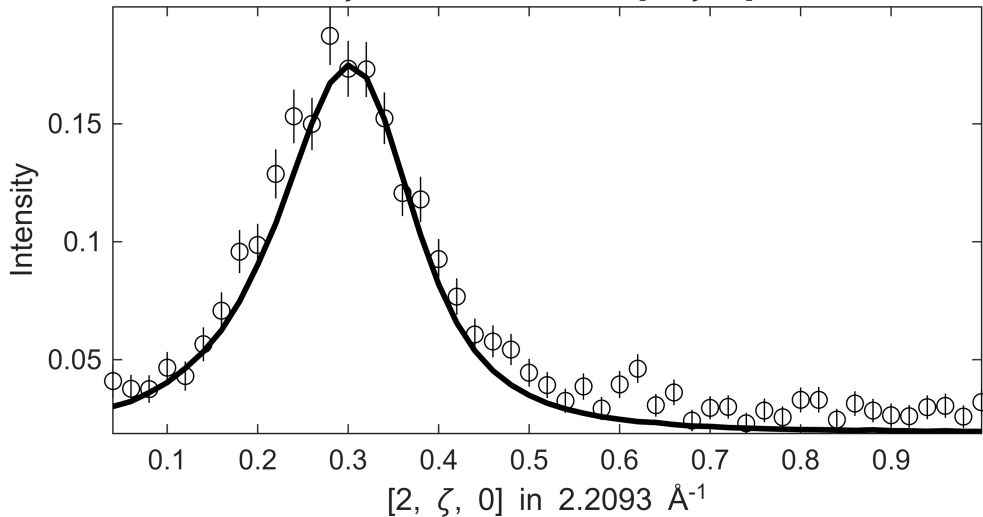
Data access time fraction: 32.3%; Transf. time frac.: 50.0%; Resulting pix preparation: 17.7%

Fe_ei200_align_sym3.sqw

S= 1 ± 0.062 ; J0= 32 ± 0.77 ; gamma= 54.4604 ± 5.26525

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $90 \leq E \leq 110$

$\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 506911 pixels -- processing data... -----> retained 460829 pixels

Number of points in input object: 506911

Fraction of object selected: 100.00% (= 506911 points) ; Access speed: 1.82GB/sec
Fraction of object retained: 90.91% (= 460829 points)

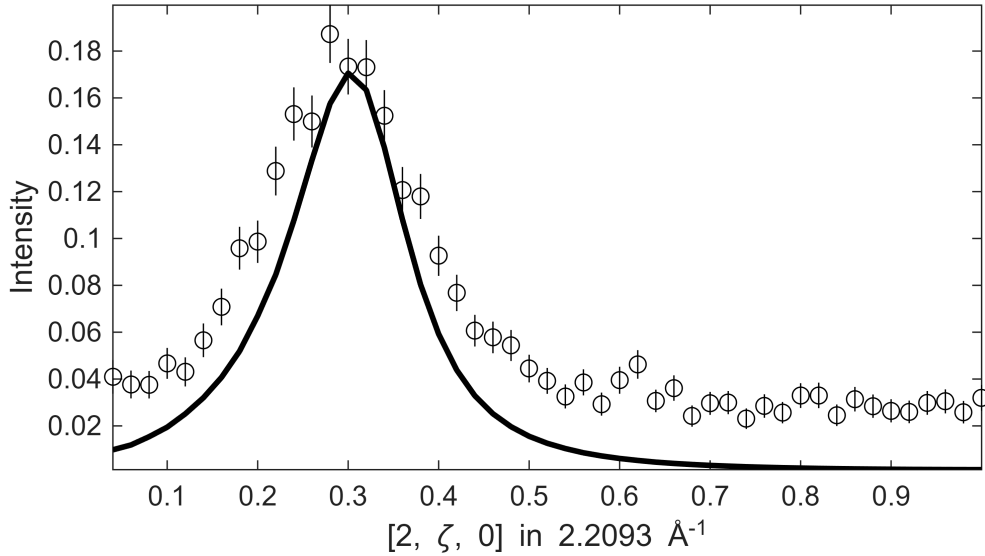
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 31.7%; Transf. time frac.: 49.5%; Resulting pix preparation: 18.9%

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $90 \leq E \leq 110$

$\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



**** En = 110±10

Warning: Dataset:12 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.139

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 636346 pixels -- processing data... -----> included 568397 pixels

Number of points in input object: 636346

Fraction of object selected: 100.00% (= 636346 points) ; Access speed: 1.89GB/sec

Fraction of object retained: 89.32% (= 568397 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 43.1%; Transf. time frac.: 54.4%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 623478 pixels -- processing data... -----> retained 557468 pixels

Number of points in input object: 623478

Fraction of object selected: 100.00% (= 623478 points) ; Access speed: 2.05GB/sec

Fraction of object retained: 89.41% (= 557468 points)

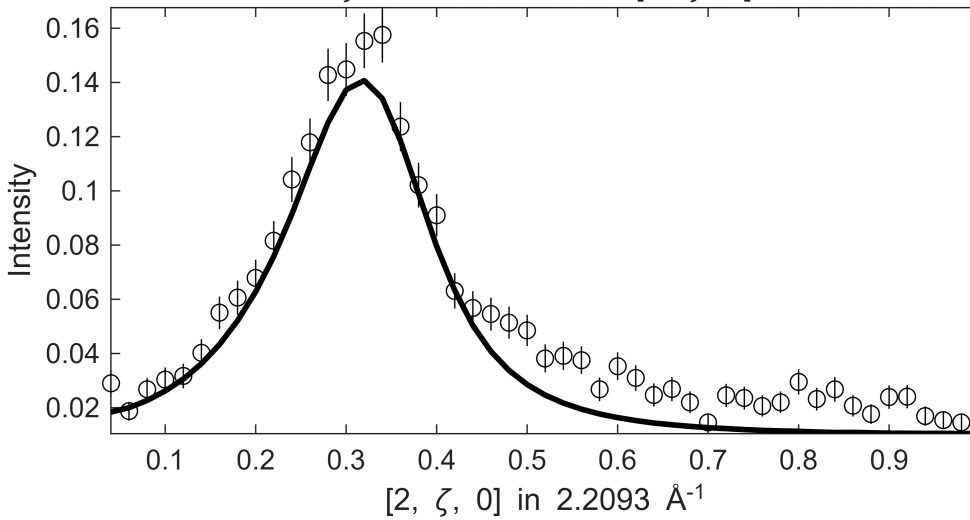
**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 28.1%; Transf. time frac.: 51.6%; Resulting pix preparation: 20.3%

Fe_ei200_align_sym3.sqw

S=0.98±0.066; J0= 33±0.88; gamma=66.6394±6.95663

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 100 ≤ E ≤ 12
ζ=-0.01:0.02:1.01 in [2, ζ, 0]

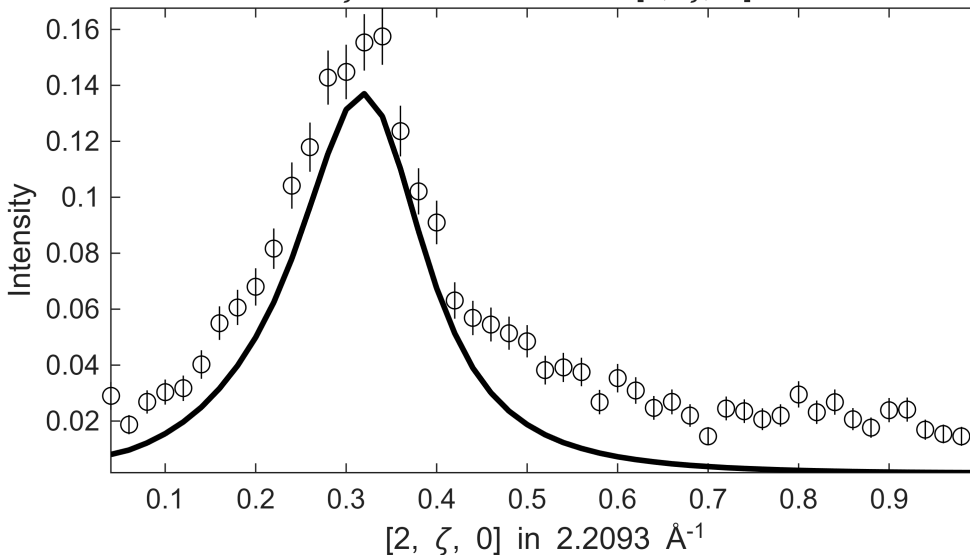


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 636346 pixels -- processing data... -----> retained 568397 pixels
-----
Number of points in input object: 636346
  Fraction of object selected: 100.00% (= 636346 points) ; Access speed: 2.03GB/sec
  Fraction of object retained: 89.32% (= 568397 points)

**** Total time in cut(sqw) : 0.1sec Including:
    Data access time fraction: 31.3%; Transf. time frac.: 46.9%; Resulting pix preparation: 21.7%
-----
```

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 100 ≤ E ≤ 12
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



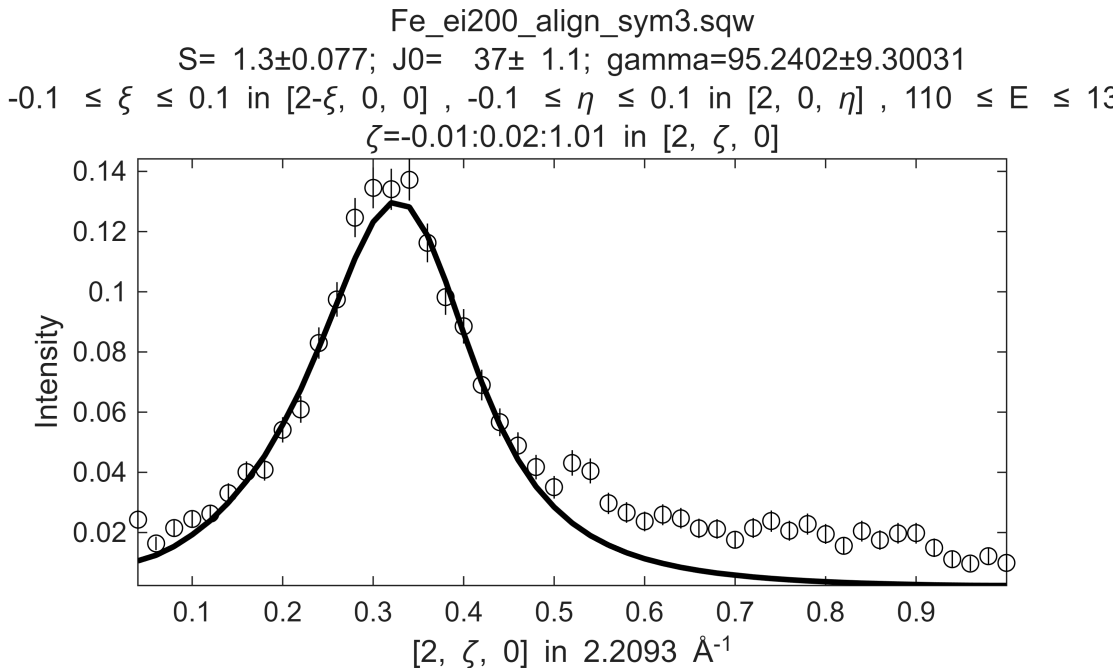
```
*****
**** En = 120±10
*****
Warning: Dataset:13 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.3173
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 1091250 pixels -- processing data... -----> included 971537 pixels
-----
Number of points in input object: 1091250
  Fraction of object selected: 100.00% (= 1091250 points) ; Access speed: 1.89GB/sec
  Fraction of object retained: 89.03% (= 971537 points)
```

```

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  53.6%; Transf. time frac.:  45.5%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 1075695 pixels -- processing data... -----> retained  956456 pixels
-----
Number of points in input object: 1075695
      Fraction of object selected:  100.00% (= 1075695 points) ; Access speed:      1.58GB/sec
      Fraction of object retained:   88.92% (=  956456 points)

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  33.6%; Transf. time frac.:  49.8%; Resulting pix preparation: 16.6%
-----

```



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 1091250 pixels -- processing data... -----> retained  971537 pixels
-----
Number of points in input object: 1091250
      Fraction of object selected:  100.00% (= 1091250 points) ; Access speed:      1.89GB/sec
      Fraction of object retained:   89.03% (=  971537 points)

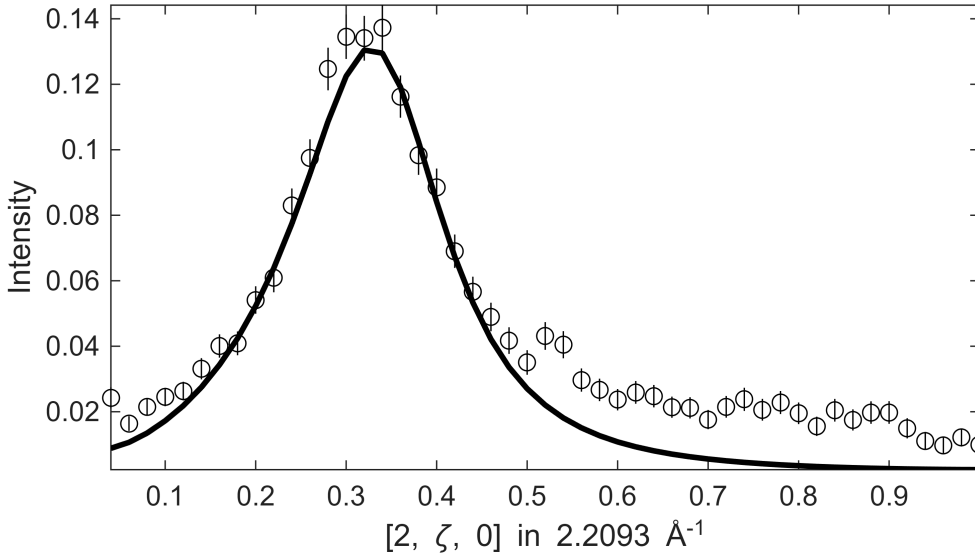
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  32.2%; Transf. time frac.:  48.2%; Resulting pix preparation: 19.6%
-----

```

Fe_ei200_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $110 \leq E \leq 130$

$\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



**** En = 130±10

Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.2478

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 2148410 pixels -- processing data... -----> included 1959539 pixels

Number of points in input object: 2148410

Fraction of object selected: 100.00% (= 2148410 points) ; Access speed: 1.56GB/sec

Fraction of object retained: 91.21% (= 1959539 points)

**** Total time in cut(sqw) : 0.1sec Including:

Data access time fraction: 65.0%; Transf. time frac.: 34.4%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 2139545 pixels -- processing data... -----> retained 1951883 pixels

Number of points in input object: 2139545

Fraction of object selected: 100.00% (= 2139545 points) ; Access speed: 1.24GB/sec

Fraction of object retained: 91.23% (= 1951883 points)

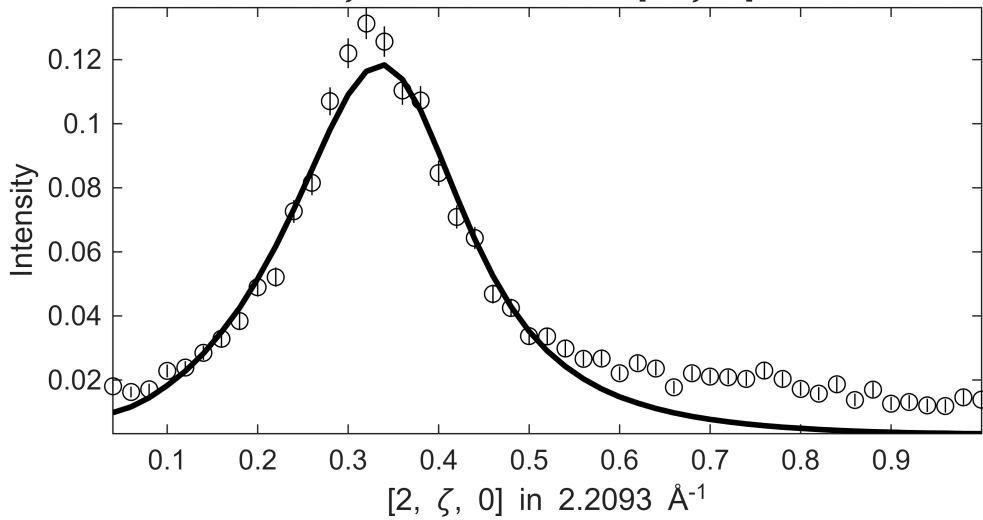
**** Total time in cut(sqw) : 0.4sec Including:

Data access time fraction: 31.2%; Transf. time frac.: 46.2%; Resulting pix preparation: 22.6%

Fe_ei200_align_sym3.sqw

S= 1.5± 0.1; J0= 40± 1.5; gamma=131.188±13.7961

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 120 ≤ E ≤ 14
ζ=-0.01:0.02:1.01 in [2, ζ, 0]

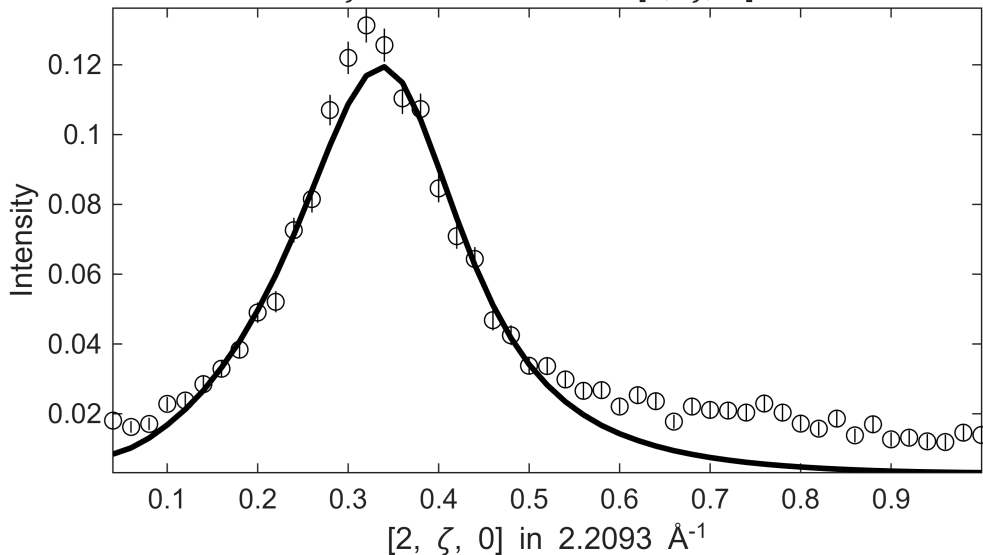


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 2148410 pixels -- processing data... -----> retained 1959539 pixels
-----
Number of points in input object: 2148410
  Fraction of object selected: 100.00% (= 2148410 points) ; Access speed: 1.40GB/sec
  Fraction of object retained: 91.21% (= 1959539 points)

**** Total time in cut(sqw) : 0.3sec Including:
      Data access time fraction: 30.4%; Transf. time frac.: 44.8%; Resulting pix preparation: 24.8%
-----
```

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 120 ≤ E ≤ 14
ζ=-0.01:0.02:1.01 in [2, ζ, 0]



```
*****
**** En = 140±10
*****
Warning: Dataset:6 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.0695
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 2322474 pixels -- processing data... -----> included 2213594 pixels
-----
Number of points in input object: 2322474
  Fraction of object selected: 100.00% (= 2322474 points) ; Access speed: 1.95GB/sec
  Fraction of object retained: 95.31% (= 2213594 points)
```

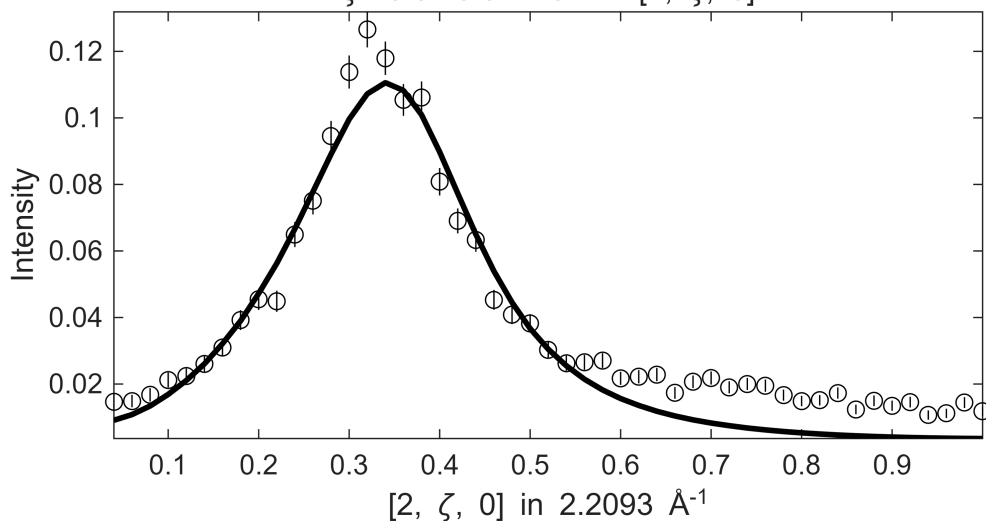
```

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction: 60.6%; Transf. time frac.: 39.0%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 2320917 pixels -- processing data... -----> retained 2212037 pixels
-----
Number of points in input object: 2320917
      Fraction of object selected: 100.00% (= 2320917 points) ; Access speed:      1.95GB/sec
      Fraction of object retained:  95.31% (= 2212037 points)

**** Total time in cut(sqw)   :      0.3sec Including:
      Data access time fraction: 31.1%; Transf. time frac.: 47.1%; Resulting pix preparation: 21.7%
-----

```

Fe_ei200_align_sym3.sqw
 $S = 1.6 \pm 0.13$; $J_0 = 41 \pm 1.9$; $\gamma = 141.539 \pm 18.1493$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $130 \leq E \leq 150$ eV
 $\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 2322474 pixels -- processing data... -----> retained 2213594 pixels
-----
Number of points in input object: 2322474
      Fraction of object selected: 100.00% (= 2322474 points) ; Access speed:      1.53GB/sec
      Fraction of object retained:  95.31% (= 2213594 points)

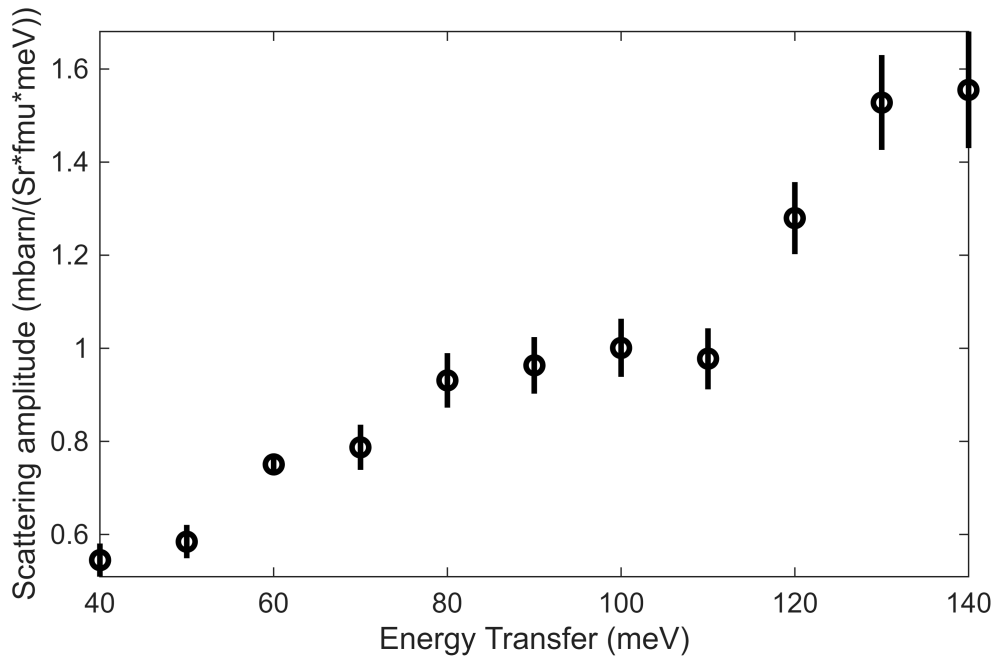
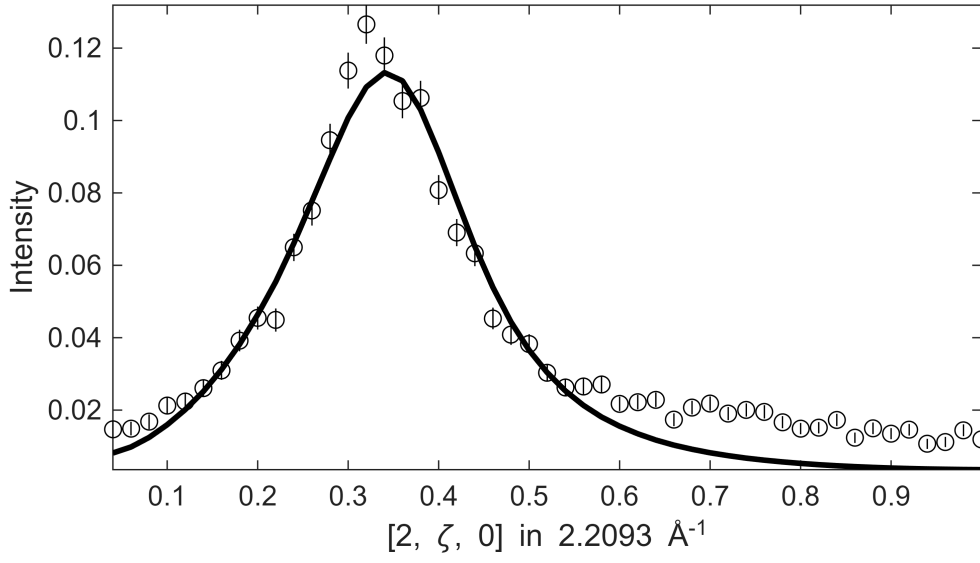
**** Total time in cut(sqw)   :      0.3sec Including:
      Data access time fraction: 36.6%; Transf. time frac.: 41.9%; Resulting pix preparation: 21.5%
-----

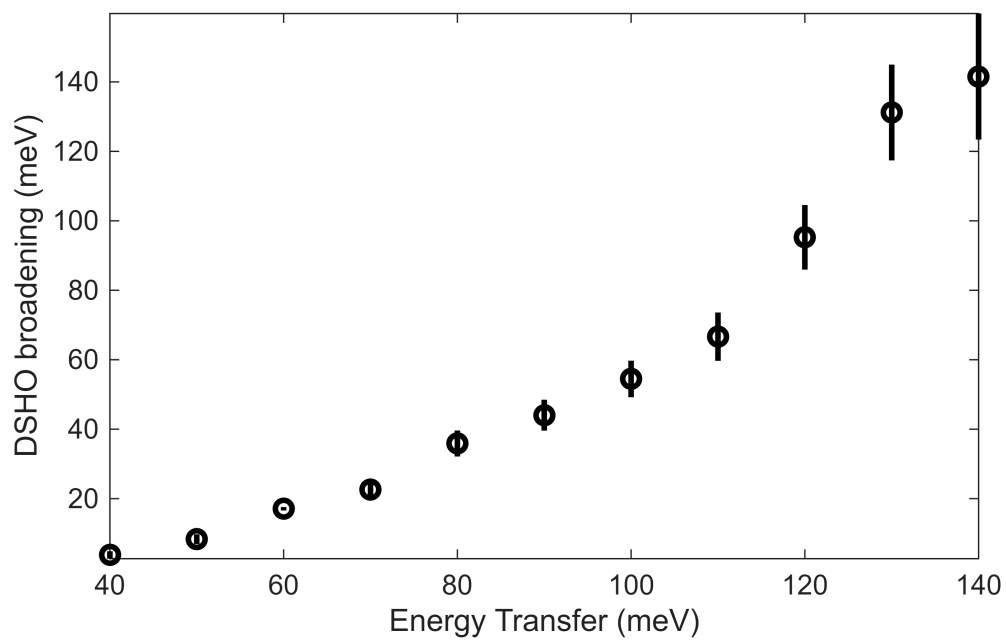
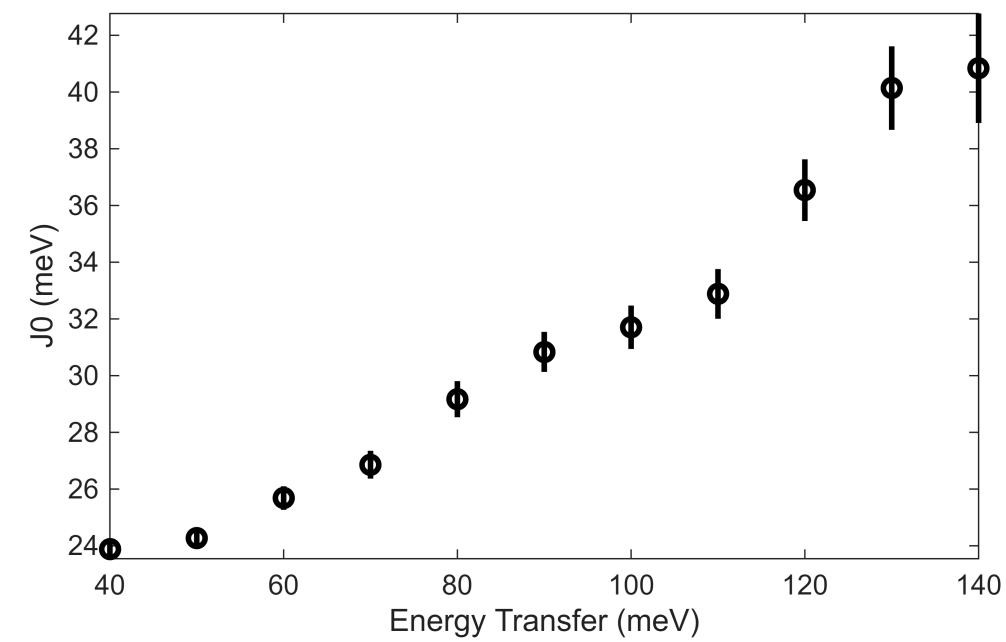
```

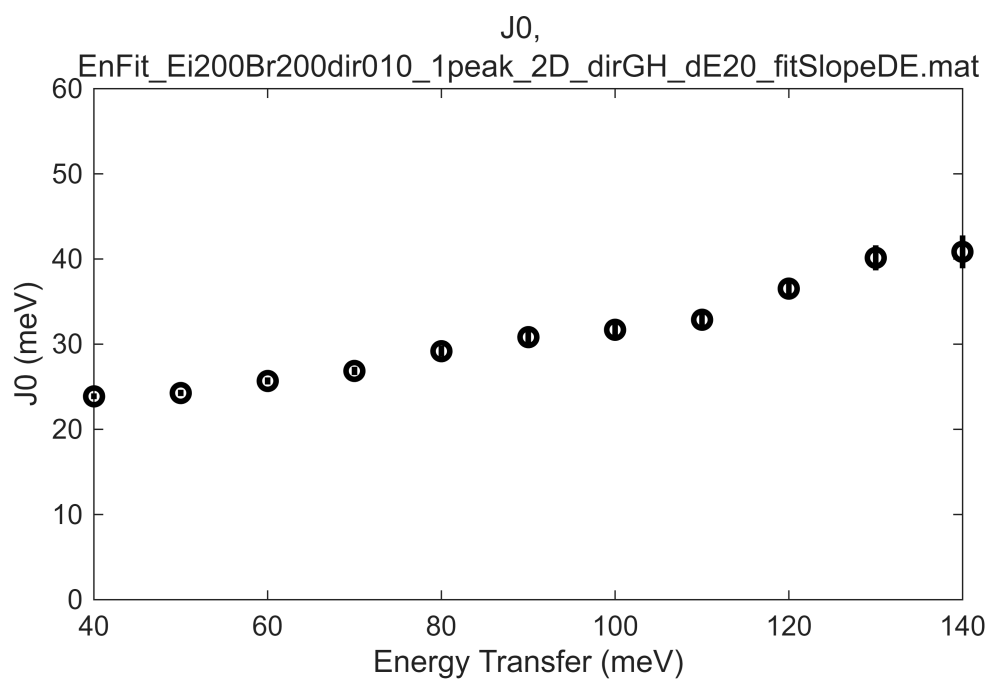
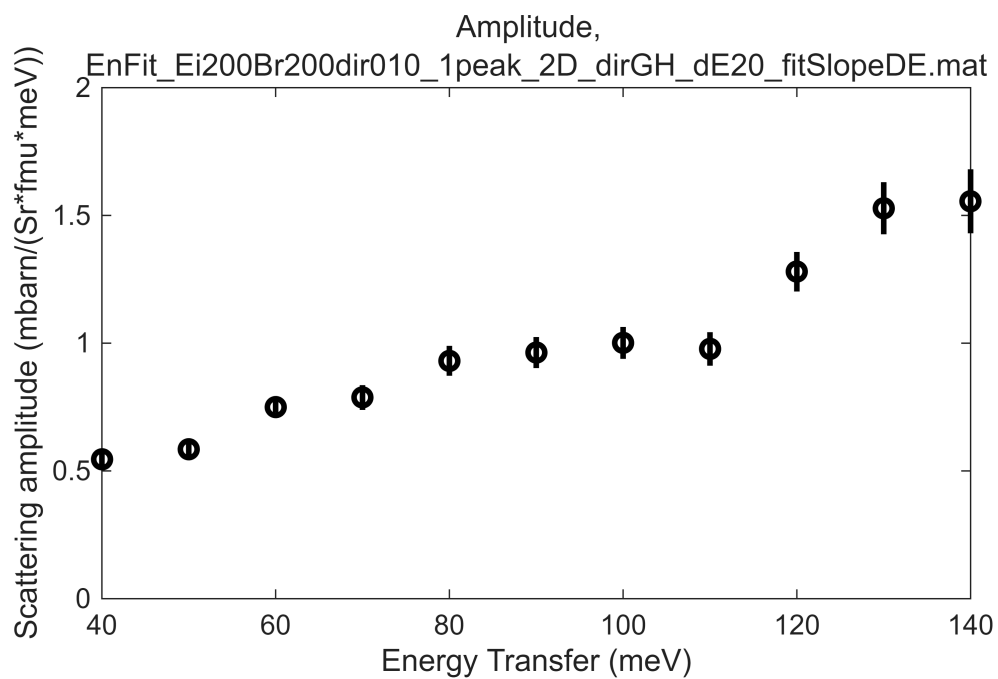
Fe_ei200_align_sym3.sqw

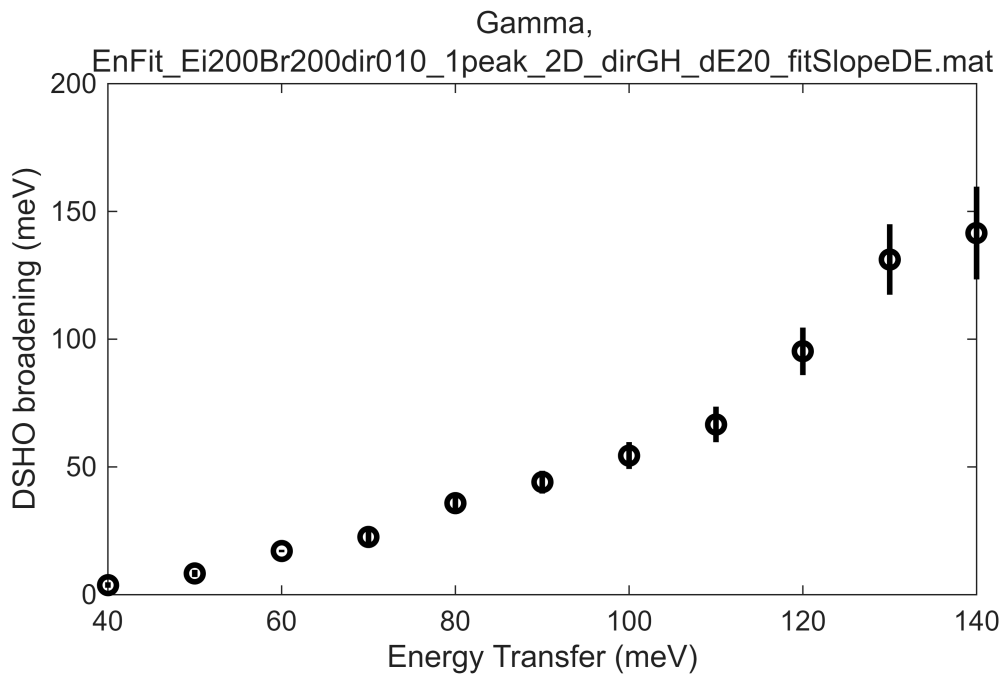
$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $130 \leq E \leq 150$ meV

$\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



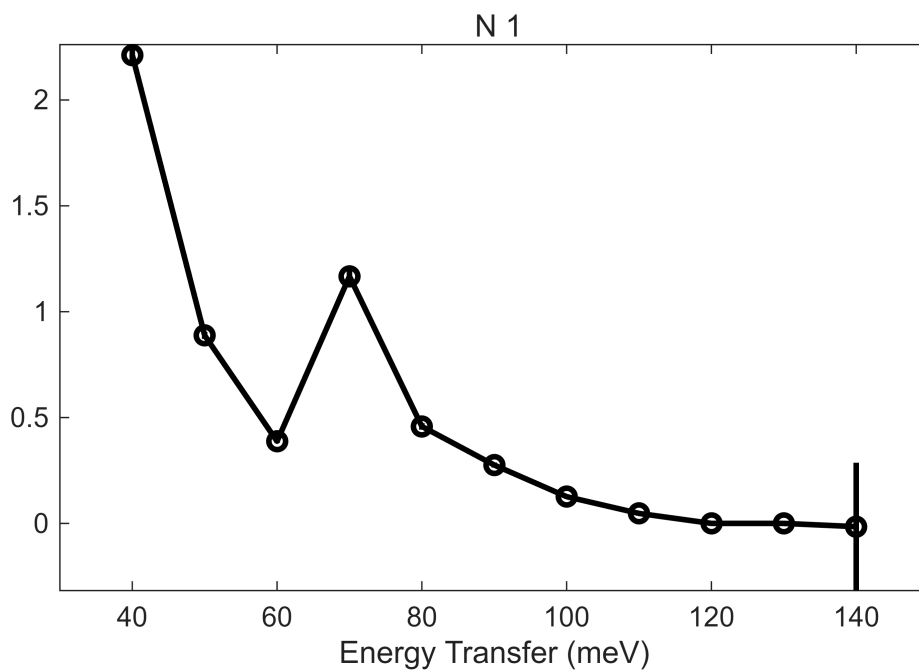




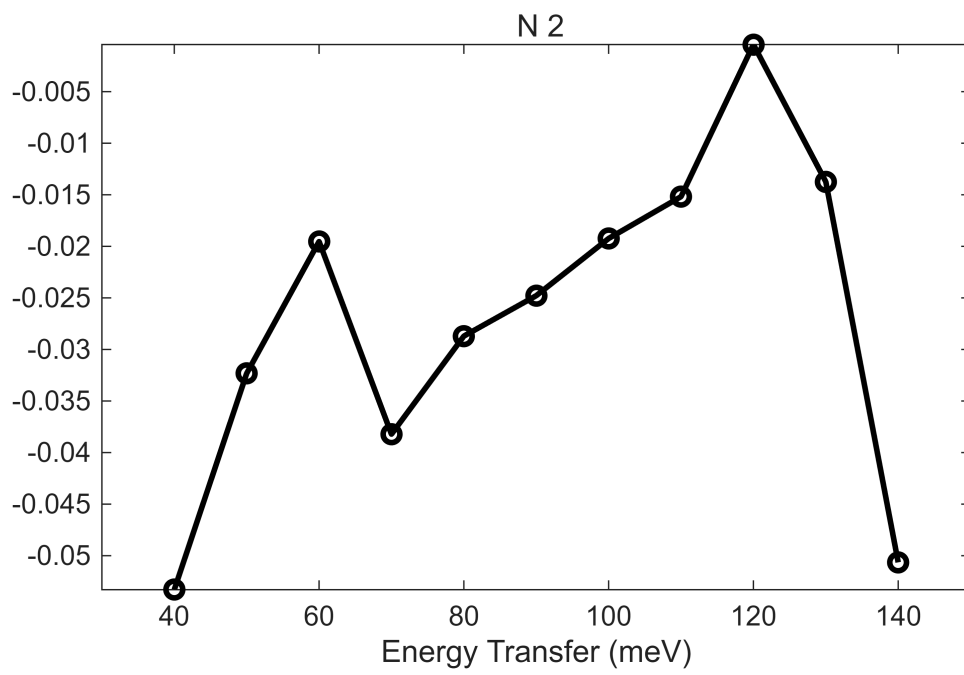


beep
beep
beep

```
bg_data = extract_bg_par(fit_res_200fix_bg.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;
```



```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure
```



```
fit_resEi200 = fit_res_200fix_bg;  
save(res_name, "fit_resEi200")
```